



Foodie traps within facebook cannabis promotional posts: Deploying multimodal deep learning AIs to monitor audience engagement

Linqi Lu^a, Xianshi Yu^b, Zhuoyan Xu^b, Hyerin Kwon^c, Akhil P. Reddy^b, Haohang Xin^c, Shifan Zhang^d, Ellie Fan Yang^e, Yin Li^b, Sijia Yang^{c,*}

^a Department of Communication, University of North Dakota, Grand Forks, USA

^b School of Computer, Data & Information Sciences, University of Wisconsin-Madison, Madison, USA

^c School of Journalism and Mass Communication, University of Wisconsin-Madison, Madison, USA

^d School of Computing, Australian National University, Canberra, Australia

^e School of Communication, Illinois State University, Normal, USA

ARTICLE INFO

Keywords:

Cannabis
Social media
Advertising
Adolescents and young adults
LLMs
Computer vision

ABSTRACT

Background: Cannabis marketing has flourished due to the legalization of recreational cannabis across multiple U. S. states, particularly on social media platforms that effectively reach adolescents and young adults. Understanding how cannabis promotional content shape audience engagement is important for monitoring exposure and informing public health and regulatory discussions.

Methods: Leveraging advanced computational analysis, including the multimodal large language model LLaVA and GPT-4o, we analyzed a comprehensive dataset of Facebook posts from 2021 to 2022 to explore associations between visual tactics and online audience engagement (e.g., likes, shares, and comments).

Results: Unsupervised clustering identified ten distinct visual themes, with human and animal portrayals, food cues being the most prevalent. High-level visual features such as food cues were consistently associated with increased shares and comments. Also, low-level color characteristics, particularly red, yellow, and orange, predicted higher audience engagement. Nearly half of posts contained health benefit-oriented messaging, whereas explicit health risk or safety information was rare.

Discussion: Visually engaging strategies in cannabis advertising on social media are associated with higher audience engagement and may increase the visibility of cannabis-related content, raising public health concerns about widespread exposure and the normalization of cannabis use.

As of 2023, 24 states and the District of Columbia have legalized recreational cannabis use, reflecting a growing acceptance of cannabis in mainstream society (National Conference of State Legislatures, 2024). This legislative change has spurred a burgeoning cannabis industry, leading to increased marketing efforts aimed at expanding the consumer base, particularly on social media platforms (Krauss et al., 2017). Adolescents and young adults experience greater exposure to cannabis-related online marketing due to their higher levels of social media use compared with other age groups (Pew Research Center, 2018). Prior research has shown that exposure to substance use content on social media is associated with increased intentions to use and actual use among youth (Primack et al., 2017). These patterns raise concerns that widespread exposure to cannabis marketing may influence cannabis

use among adolescents and young adults, with potential public health implications.

While researchers have analyzed cannabis-related online marketing content, efforts are often limited to manual coding of selected brands and locations (Cui et al., 2024; Luc et al., 2020; Spillane et al., 2021). Recent computational analyses of cannabis social media posts have emerged using machine learning but typically focus only on textual data while largely neglecting the rich visual content (Allem et al., 2020; Deng et al., 2024; Hu et al., 2021; Lu et al., 2025; Yuan et al., 2024). Yet marketing research shows that visuals can simulate experiences, represent brand values, enhance emotional reactions and consumer-brand connections (Holbrook and Batra, 1987; Lang et al., 1993), improve memory recall (Goldberg and Gorn, 1987), and drive audience

* Correspondence to: School of Journalism and Mass Communication, University of Wisconsin-Madison, Mail: 5160 Vilas Hall, 821 University Avenue, Madison, WI 53706, USA.

E-mail address: sijia.yang@alumni.upenn.edu (S. Yang).

<https://doi.org/10.1016/j.drugalcdep.2026.113128>

Received 7 August 2025; Received in revised form 6 March 2026; Accepted 10 March 2026

Available online 14 March 2026

0376-8716/© 2026 Elsevier B.V. All rights reserved, including those for text and data mining, AI training, and similar technologies.

engagement (Lang et al., 1993; Pieters et al., 2010). In cannabis marketing specifically, online advertising increasingly uses youth-targeted visual appeals, including imagery emphasizing sensory attributes (e.g., bright colors, fruity flavors) and food cues (e.g., portraying edibles as candies and chocolates) that align with youth preferences (Moreno et al., 2019).

To monitor the cannabis industry's online advertising, researchers need to develop capabilities to analyze large-scale multimodal posts promoting recreational cannabis products. Such data encompasses both low-level textual and aesthetic features (brightness, contrast, colors) and high-level semantic features (topics, food cues, object portrayals) (Peng, 2022). In this study, we present a computational analysis of Facebook cannabis commercial posts collected between 2021 and 2022, using a human-AI hybrid approach that combines unsupervised image clustering with supervised extraction of theoretically relevant visual features. Our approach builds on recent advances in multimodal Large Language Models (LLMs) like LLaVA and GPT-4o, which enhance capabilities for processing rich multimodal social media content (Peng et al., 2024; Radford et al., 2021). The first goal of this study is to identify and characterize the categories of visual features present in cannabis-related Facebook promotional posts. Specifically, we explore the visual feature categories that appear most frequently across the corpus, as identified through unsupervised clustering and supervised classification. Accordingly, we ask:

RQ1: (a) What categories of visual features are present in cannabis-related Facebook promotional posts? and (b) what are the predominant visual appeals?

Our second goal is to systematically identify visual features (i.e., both lower-level aesthetic attributes and higher-level semantic appeals) that predict online engagement after accounting for the influences of text and post-level metadata. Online engagement, measured through likes, comments, and shares, signals audience interest and often serves as the primary metric that social media recommendation algorithms aim to maximize (Bakshy et al., 2015). The more people like, comment, and share a post the more ranking algorithms prioritize it in search results leading to a reinforcing cycle of exposure for popular posts (Cappella et al., 2014; Thorson and Wells, 2016). Therefore, identifying visual features that predict audience engagement with marketing posts can inform both ongoing discussions about regulating cannabis advertising and campaigns to educate adolescents and young adults about cannabis products and use—whether to communicate overlooked health risks of recreational cannabis use or correct unverified health benefit claims (Johnston et al., 2020; Cavazos-Rehg et al., 2019; Ishida et al., 2020).

RQ2a-c: Which automatically extracted aesthetic features (e.g., specific colors, colorfulness, hue count) in cannabis promotional posts will predict higher levels of audience engagement, in terms of (a) shares, (b) comments, and (c) likes?

H1a-c: The use of food cues predicts increased audience engagement in terms of (a) shares, (b) comments, and (c) likes.

H2a-c: Portrayals of humans predicts increased audience engagement in terms of (a) shares, (b) comments, and (c) likes.

H3a-c: Portrayals of animals predicts increased audience engagement in terms of (a) shares, (b) comments, and (c) likes.

RQ3: (a) To what extent do Facebook cannabis promotional posts mention health benefits and health risks? (b) What health-related themes are present in these posts?

1. Methods

1.1. Data collection and screening

Data of Facebook posts were collected. A pre-established list of search keywords (e.g., cannabis, edibles, kush, marijuana, see Appendix II for a full list) were used to retrieve a comprehensive initial corpus of 171,917 Facebook posts from the CrowdTangle API, covering a two-year period from January 1, 2021 to December 31, 2022. CrowdTangle

additionally provides aggregated engagement metrics including the total number of likes, shares, and comments at the time of data retrieval, along with meta-data such as date, message, and account follower numbers, etc. Images including those in embedded URLs, text, engagement metrics, and meta-data were scraped and downloaded for all posts.

A two-stage automated screening was conducted to select posts related to cannabis promotional posts. First, to identify posts related to cannabis products, a pre-trained multimodal deep learning model LLaVA (Liu et al., 2023) was employed to analyze post data. LLaVA (LLaVA-v1.6-Vicuna-13b), publicly available at Hugging Face (huggingface.co), has the capability to jointly analyze images and text inputs, well suited for this task. Image and text in a post were examined by LLaVA, which returned a binary label ([Yes/No]) to classify whether the post is related to cannabis products. Results were verified manually based on a subset of data. LLaVA identified 145,140 posts as cannabis-related after deduplication. We validated classification accuracy by comparing classifier labels with manual coding on a random subsample ($n = 500$), achieving 94% agreement. Second, to further select posts that promote cannabis products, a large language model (GPT-4o) was employed to examine text data from individual posts. The account name, account page description, and text in the post were input to GPT-4o, which then classified the post as commercial or non-commercial. GPT-4o was prompted to consider promotional posts defined by characteristics such as product mentions, sales-oriented language, and service offerings. This classification resulted in a final analytical sample of 27,242 posts promoting cannabis products. Inter-coder reliability was assessed with three coders on 200 random subsamples (Krippendorff's alpha = 0.89, agreement = 92%). Overall workflow was presented in Fig. 1.

1.2. Visual features extraction

Low-level aesthetic features. Low-level, color-related features, including colorfulness, hue count, and the presence of specific colors (red, orange, and yellow), were extracted from images using *Athec* (Peng, 2022) a Python library designed for computational aesthetics. Colorfulness is calculated using the pixel distribution in the opponent color space. It has been found to highly correlate with human rating in a psychophysical experiment (Hasler and Suesstrunk, 2003). Hue count is simply calculated by assigning each pixel into one of 20 hue bins and counting the number of bins with a significant portion of pixels.

High-level semantic features. High-level, object-related features, such as the presence of various food cues (e.g., portrayals of candies, bakery items, chocolate, and fruit), human, animal, and were also extracted from images using LLaVA. The resulting features were further verified manually using a subset of the data. Specifically, LLaVA was prompted to classify whether a specific cue was present in an image. For example, to detect the portrayals of “bakery,” the model received the text prompt “Determine if the product contains portrayals of a bakery item, such as a cookie, brownie, cake, etc.” (see Appendix III). The image was processed alongside this prompt, and LLaVA output a binary label indicating the presence or absence of “bakery”. To evaluate LLaVA's performance, we compared model-generated classification with human annotations of a randomly selected subsample ($n = 100$). The LLaVA model reached an acceptable level of agreement across all targeted features (F1 scores $>.91$, Krippendorff's alphas $>.86$). Among all high-level semantic features, branding imagery was most prevalent (64.8%) in commercial posts, followed by portrayals of humans (29.9%), food cues (17.9%), weed plants (11.4%), the use of animation (4.2%), and lastly, depictions of animals (2.6%).

1.3. Clustering and statistical analyses

Image clustering and qualitative analysis. An unsupervised machine learning approach was implemented to identify the underlying themes within our unlabeled image datasets. This approach integrated deep

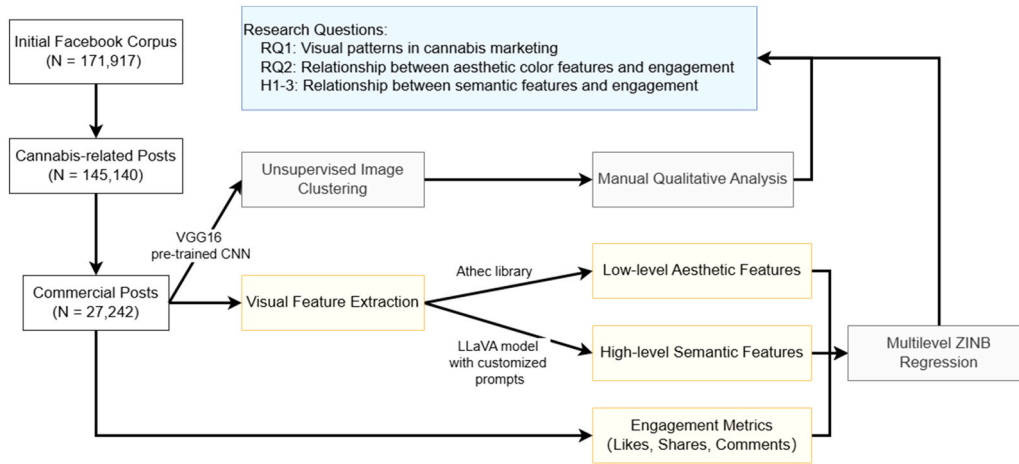


Fig. 1. Workflow.

learning models in the form of convolutional neural networks (CNNs) and clustering algorithms. Specifically, the VGG16 model (Simonyan and Zisserman, 2014), pretrained on the ImageNet dataset, was used to extract visual features representing each image. VGG16 has been proved to provide rich, high-dimensional feature vectors that effectively capture complex image content. Principal component analysis (PCA) was then performed on these high-dimensional features and principal components that accounted for at least 80% of the variance were retained. The K-means clustering algorithm was further applied to lower-dimensional features, aiming to group images based on their themes (MacQueen, 1967). To determine the optimal number of clusters, the K values were varied from 2 to 15 and examined using the elbow method and silhouette analysis, ensuring that the chosen K provided meaningful and interpretable groupings (Ketchen and Shook, 1996). Based on the optimal number of clusters (K=10), 100 example images were sampled from each cluster and manually inspected to provide a semantic interpretation.

Modeling the relationships between visual features and engagement metrics. Multilevel random intercepts zero-inflated negative binomial (ZINB) regression models were employed to predict the counts of likes, shares, and comments, based on low-level and high-level image features, as well as control variables. This statistical approach was selected because engagement metrics count variables that exhibit overdispersion and a high frequency of zero counts (Atkins et al., 2013; Hilbe, 2011). Furthermore, the inclusion of random intercepts at the account level accounts for potential dependencies between observations where posts were nested under account (Raudenbush and Bryk, 2002). Input features of interest including both color-related aesthetic features and semantic features, along with control variables (e.g., textual features, account profile characteristics, time stamp of posting), were modeled as fixed effects. Multicollinearity was benign (VIFs < 2.3, see Appendix I).

1.4. Thematic analysis of health messaging

To address RQ3, we conducted a qualitative thematic analysis to examine health-related claims and themes in cannabis commercial posts, including both claims about health benefits and those about health risks or safety concerns. AI-assisted tools were used solely to facilitate text processing and support the identification of thematic clusters, while theme definition, interpretation, and selection of illustrative examples were conducted by our team, in line with prior mixed-methods research (Grimmer et al., 2021; Lu et al., 2025).

2. Results

2.1. Unsupervised image clustering analyses

Human interpretation of the unsupervised clustering results is summarized in Table 1 and Fig. 2. Among the ten image clusters, the most frequently deployed visual feature was portrayals of humans and animals. Evidence also showed that deploying food cues to highlight sensory attributes was a prominent strategy. Not surprisingly, images featuring brands and products constituted large shares of the total corpus. Consistent with the model's capacity to process multimodal data, clusters characterized by textual features, such as those emphasizing "healthy" lifestyles and wellness claims about CBD oils, also emerged. Interestingly, two clusters primarily featured spreadsheets and infographics, suggesting the popularity of using data visualization in online cannabis marketing.

2.2. Visual features predicting engagement

Regarding which color-related aesthetic features predicted engagement (RQ2), results from the random intercepts multilevel ZINB models indicated that hue count positively predicted likes (IRR = 1.01, 95% CI [1.01, 1.02], $\beta = 0.01$, SE = 0.003, $p < .001$), while colorfulness showed a small but significant negative association (IRR = 1.00, 95% CI [1.00,

Table 1
Summary of Image Clusters and Visual Content Categories (N = 27,242).

Content Category	Explanation	Count
Promotion Using Human and Animal	Posts featuring people or animals to promote cannabis-related content.	8154 (29.9%)
Ads Using Healthy Concept	Advertisements promoting cannabis through health-related claims or wellness themes.	5320 (19.5%)
Promotion Using Food Cue	Marketing that associates cannabis with food, flavors, or edibles.	4141 (15.2%)
Cannabis-Related Branding	Posts showcasing brand logos, packaging, or promotional branding elements.	2997 (11.0%)
Product Display (CBD Oil)	Posts showcasing brand logos, packaging, or promotional branding elements.	2582 (9.5%)
CBD Health and Wellness Marketing	Posts marketing CBD for therapeutic, relaxation, or self-care purposes.	2378 (8.7%)
Product Display	General product displays beyond CBD oil, including cannabis accessories.	1119 (4.1%)
Infographics	Visual content providing educational or informational cannabis-related insights.	297 (1.1%)
Spreadsheet	Visual content providing educational or informational cannabis-related insights.	130 (0.5%)
Retail Display	Posts showcasing cannabis products in retail settings, such as dispensaries.	124 (0.5%)

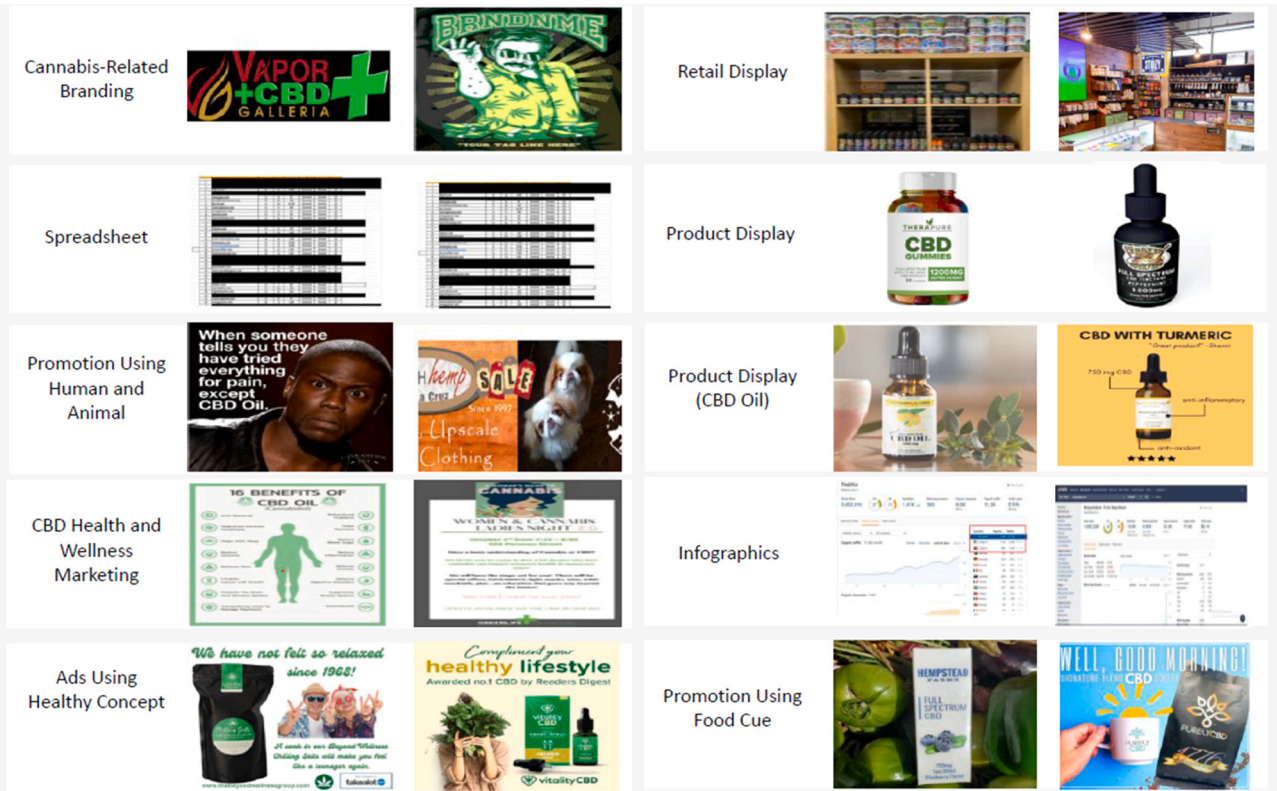


Fig. 2. Content Categories and Examples.

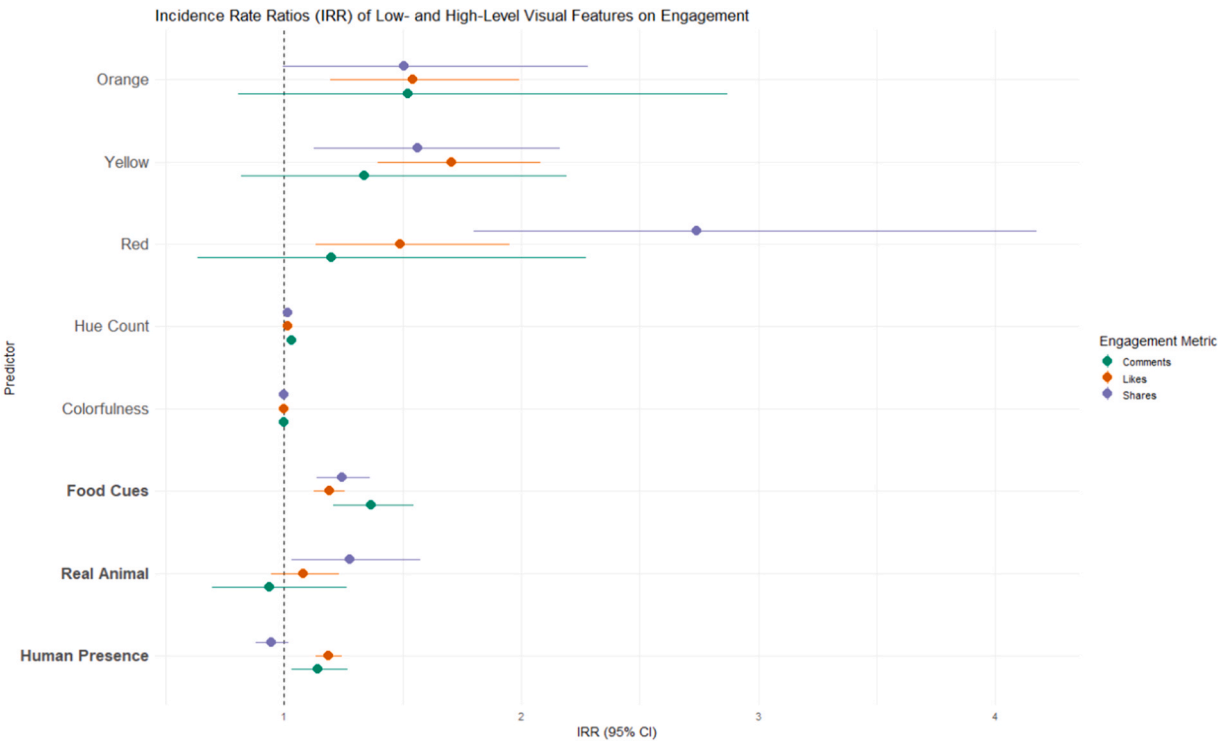


Fig. 3. Estimates for Engagement (Likes, Comments, Shares), Note. Points represent incidence rate ratios (IRRs) estimated from multilevel random-intercept zero-inflated negative binomial (ZINB) models predicting social media engagement (likes, comments, shares). Error bars indicate 95% confidence intervals on the IRR scale (exponentiated log-link coefficients). The dashed horizontal line at IRR = 1 denotes no association. Models include random intercepts at the account level and adjust for visual semantic features, color-related aesthetic features, account subscriber count (log-transformed), posting day (weekend vs. weekday), and other covariates. High-level semantic features (bolded) were classified using LLaVA; low-level color features were extracted using computer vision techniques.

1.00], $\beta = -0.001$, $SE = 0.0005$, $p = .03$). The specific color red ($IRR = 1.49$, 95% CI [1.13, 1.95], $\beta = 0.40$, $SE = 0.139$, $p = .004$), yellow ($IRR = 1.71$, 95% CI [1.40, 2.08], $\beta = 0.53$, $SE = 0.102$, $p < .001$), and orange ($IRR = 1.54$, 95% CI [1.20, 1.99], $\beta = 0.43$, $SE = 0.13$, $p = .001$) were positively associated with likes. Regarding shares, hue count ($IRR = 1.02$, 95% CI [1.01, 1.03], $\beta = 0.02$, $SE = 0.01$, $p = .002$) and specific colors including red ($IRR = 2.74$, 95% CI [1.80, 4.18], $\beta = 1.01$, $SE = 0.22$, $p < .001$) and yellow ($IRR = 1.56$, 95% CI [1.13, 2.17], $\beta = 0.45$, $SE = 0.167$, $p = .008$) were positive predictors. Lastly, among color-related features, hue count was the only consistent positive predictor of comments ($IRR = 1.03$, 95% CI [1.02, 1.05], $\beta = 0.03$, $SE = 0.01$, $p < .001$).

H1–3 concern the association between high-level semantic features and engagement metrics. First, results from the LLaVA model for visual feature classification produced the following descriptive distribution: branding imagery was most prevalent (64.8%) in promotional posts, followed by portrayals of humans (29.9%), food cues (17.9%), weed plants (11.4%), the use of animation (4.2%), and lastly, depictions of animals (2.6%). Second, the multilevel random intercepts ZINB model showed that food cues consistently predicted higher shares ($IRR = 1.24$, 95% CI [1.14, 1.36], $\beta = 0.22$, $SE = 0.05$, $p < .001$), comments ($IRR = 1.37$, 95% CI [1.21, 1.54], $\beta = 0.31$, $SE = 0.06$, $p < .001$), and likes ($IRR = 1.19$, 95% CI [1.13, 1.24], $\beta = 0.17$, $SE = 0.03$, $p < .001$). The effects of portrayals of humans and animals were less stable. Whereas depictions of humans predicted more likes ($IRR = 1.19$, 95% CI [1.13, 1.24], $\beta = 0.17$, $SE = 0.03$, $p < .001$) and comments ($IRR = 1.14$, 95% CI [1.03, 1.27], $\beta = 0.13$, $SE = 0.05$, $p = .01$) but not shares, portrayals of animals only predicted more shares ($IRR = 1.28$, 95% CI [1.03, 1.58], $\beta = 0.24$, $SE = 0.11$, $p = .02$) among these engagement metrics. These results suggest that food cues are associated with increased audience engagements, whereas portrayals of humans or animals may not have a similar effect. H1 and H2 received partial support. See Table 2 for the full model results.

2.3. Prevalence of health-related claims in cannabis promotional posts

To address RQ3(a), we assessed the prevalence of health-related claims in the textual component of 26,471 unique Facebook cannabis promotional posts. More than half of posts (51.27%, $n = 13,572$) mentioned no identifiable health-related claims. Nearly half of the posts

(47.92%, $n = 12,685$) included language promoting health benefits, such as references to wellness, relaxation, or perceived therapeutic effects, without accompanying risks or safety information. By contrast, mentioning health risks or safety concerns was rare. Only 0.80% ($n = 213$) of posts included information on health risks or safety concerns in the absence of mentioning any health benefits, and almost no posts mentioned both health benefits and health risks.

To address RQ3(b), we conducted a qualitative thematic analysis to unpack claims related to both health benefits and health risks (see Table 3). Among posts mentioning health benefits ($N = 12,685$), the most prevalent theme framed cannabis use as part of general wellness and lifestyle enhancement, positioning cannabis products within routines of self-care, balance, or everyday well-being. These messages frequently employed broad, affirmative language emphasizing lifestyle optimization and personal wellness. A substantial subset of posts emphasized CBD-related therapeutic benefits, such as relaxation, healing, balance, or recovery. Additional themes emphasized stress reduction, anxiety relief, and relaxation support, frequently portraying cannabis use as a means of emotional regulation to cope with daily stressors, highlighting benefits for mood regulation or calming effects for both humans and animals. Themes related to sleep and recovery were also evident, with posts highlighting improved sleep quality, nighttime rest, or recovery benefits, often tied to specific product formulations such as gummies or melatonin-infused products. Less frequently, posts referenced pain and physical relief, typically using language to suggest relief from soreness, inflammation, or general discomfort.

In contrast, posts warning about health risks or safety concerns were comparatively rarer ($N = 213$) and limited in scope. The most common risk-related posts included age restrictions and legal compliance statements, such as references to adult-use requirements, licensing, and regulatory disclaimers. A smaller subset of posts included responsible-use and impairment warnings, typically discouraging driving or operating machinery under the influence of cannabis. These warnings were often brief and embedded within longer promotional messages. Pregnancy and reproductive cautions were present but infrequent, generally limited to short warnings against use during pregnancy or breastfeeding. Some posts included general health risk or safety warnings, occasionally listing multiple potential risks such as impairment, habit formation, or legal restrictions. These messages were typically presented after the long

Table 2
Multilevel Zero-Inflated Negative Binomial Regression Predicting User Engagement.

Predictor	Likes		Comments		Shares	
Visual Features	IRR	CI	IRR	CI	IRR	CI
Food cues	1.19 ***	[1.13,1.26]	1.37 ***	[1.21,1.54]	1.24 ***	[1.14,1.36]
Human presence	1.19 ***	[1.13,1.24]	1.14 *	[1.03,1.27]	0.95	[1.09,1.36]
Real animal	1.08	[0.94,1.23]	0.94	[0.70,1.26]	1.28 *	[1.03,1.58]
Color Features						
Colorfulness	0.99 *	[1.00,1.00]	0.99	[1.00,1.00]	1.00	[1.00,1.00]
Hue count	1.01 ***	[1.01,1.02]	1.03 ***	[1.02,1.05]	1.02 **	[1.01,1.03]
Red	1.49 **	[1.13,1.95]	1.20	[0.63,2.27]	2.74 ***	[1.80,4.18]
Yellow	1.71 ***	[1.40,2.08]	1.34	[0.82,2.19]	1.56 **	[1.13,2.17]
Orange	1.54 ***	[1.20,1.99]	1.52	[0.81,2.87]	1.51 *	[0.99,2.28]
Covariates						
Weed plant	1.07 *	[1.00,1.15]	1.07	[0.92,1.24]	1.22 ***	[1.09,1.36]
Animation	0.93	[0.84,1.02]	0.92	[0.73,1.16]	1.05	[0.90,1.15]
Branding	0.99	[0.94, 1.03]	1.00	[0.90, 1.11]	1.06	[0.99, 1.15]
Call to action	0.87 ***	[0.83,0.92]	0.99	[0.89,1.11]	1.09 *	[1.00,1.19]
Health benefit	0.97	[0.93,1.02]	0.77 ***	[0.69,0.85]	0.90 *	[0.83,0.98]
Health risk	1.16	[0.86,1.56]	2.05 *	[1.13,3.73]	2.00 **	[1.22,3.29]
Warning label	0.75	[0.56,1.02]	0.72	[0.37,1.43]	0.54 *	[0.33,0.89]
Account subscriber	1.45 ***	[1.42,1.49]	1.80 ***	[1.74,1.86]	1.34 ***	[1.30,1.39]
Weekend post	0.92 **	[0.88,0.97]	0.88 *	[0.79,0.98]	0.85 ***	[0.78,0.92]

Note. alues are incidence rate ratios (IRRs) from multilevel random-intercept zero-inflated negative binomial regression models with random intercepts ($N = 27,242$ posts nested within 11,649 accounts). IRRs greater than 1 indicate higher expected engagement counts, whereas IRRs less than 1 indicate lower expected counts. Bracketed values denote 95% confidence intervals. Statistical significance is determined by whether the confidence interval excludes 1. Significance levels: $p < .001$ ***, $p < .01$ **, $p < .05$.*

Table 3
Health-Related Textual Claims in Cannabis Promotional Posts.

Message Type	Theme	Description	Examples
Health Benefits (N = 12,685)	Wellness and lifestyle enhancement	Framing cannabis use as part of general wellness, self-care, balance, or lifestyle optimization.	"Incorporate our products into your daily wellness routine and unwind naturally." "Happy Father's Day in appreciate of all the wonderful dads out there we will be doing medical marijuana certification for only \$42.00 one day only so make sure you let all the cool dads out there know just click this link now" "Our CBD oil supports relaxation and everyday balance." "Introducing our product of the month: R+CO Super Garden CBD for healing yourself... Stop by Jenny's Salon to get some for yourself!" "Our body shapers are made with therapeutic, healing, and refreshing components such as CANNABIS"
	CBD-related therapeutic framing	Referencing perceived therapeutic or wellness benefits of CBD products, such as stress relief, relaxation, sleep support, or recovery.	"Take the edge off and relax after a long day." "Try my delicious Blessed Biscuits... Anxiety Stress Inflammation Flow, So Mote it Be..." "Did you know that CBD helps nervous pups? Improves skin health? Helps with seasonal allergies? Promotes healthy joints? Now you know! Time to shop our CBD chews and more at lolahemp.com..."
	Stress, anxiety, and relaxation support	Emphasizing emotional relief, calmness, or stress reduction associated with cannabis use.	"Wind down at night and enjoy more restful sleep." "Looking for a deeper sleep and easier days? Our 10 mg Gummy Bears come in original CBD for gentle daytime dosing, and CBD + 5 mg of Melatonin for more restful nights. Thanks, @hey_itslori for the beautiful photo!"
	Sleep and recovery narratives	Highlighting improved sleep, rest, or nighttime recovery as outcomes of cannabis use.	"Perfect for soothing sore muscles after a workout." "In the United States, an average of 655,000 people die each year... CHANGING LIVES ONE DROP AT A TIME CBD HEART HEALTH"
Health Risks & Safety (N = 213)	Pain and physical relief	Mentioning relief from discomfort, soreness, or general physical pain.	"...For adult use only. Must be 21 +." "...Come get the goods early! Please consume responsibly. For use only by 21 years of age or
	Age restrictions and legal compliance	Referencing legal age limits, licensing status, or regulatory compliance, functioning	

Table 3 (continued)

Message Type	Theme	Description	Examples
		primarily as disclaimers.	older. Keep out of the reach of children. Marijuana can impair concentration, coordination, and judgment. Do not operate a vehicle or machinery under the influence of this drug." "Do not operate a vehicle or machinery under the influence of this drug. For use only by adults twenty-one years of age and older. Keep out of the reach of children..." #MuleExtracts #MuleCarts #takethehighroad #oregonedibles"
	Responsible use and impairment warnings	Brief warnings discouraging driving or operating machinery after cannabis use.	"Please consume responsibly." "Slow down this weekend with our Pineapple Haze + Banana Kush cartridges! Please use responsibly & stay safe!" "Effects may vary. Use responsibly." "Not recommended for use during pregnancy."
	Pregnancy and reproductive cautions	Warning against use during pregnancy or breastfeeding.	
	General health risk or safety warnings	Generic references to potential risks.	"I cannot believe we are the FIRST to put a THC Bloody Mary drink on the Legal Market! ... (1) Marijuana has intoxicating effects and may be habit forming and addictive; (2) Marijuana impairs concentration, coordination, and judgement. Do not operate a vehicle or machinery under its influence; (3) There are health risks associated with consumption of marijuana; (4) For use only by adults twenty-one and older. Keep out of the reach of children; (5) Marijuana should not be used by women who are pregnant or breast feeding. License 3b-23468" "Today is Medible Monday at Edmore Provisionary. Save 15% off any edible. http://agrimedmi.com/edmore-provisionary/? **For use by individuals 21 years of age or older only. Keep out of reach of children. It is illegal to drive a motor vehicle while under the influence of marijuana. National Poison Control Center

(continued on next page)

Table 3 (continued)

Message Type	Theme	Description	Examples
			1-800-222-1222. Sponsored by Edmore Provisionary"

promotion messages.

3. Discussion

Given the growing cannabis industry’s increasingly online marketing (Trangenstein et al., 2019; Trangenstein et al., 2022), there is an urgent need to monitor both textual and visual features deployed in online promotional posts, particularly those appealing to adolescents and young adults. This study demonstrates the utility of a computational approach that combines start-of-the-art deep learning computer vision techniques (i.e., VGG16 CNN, LLaVA, and GPT-4o) with researcher expertise to efficiently identify both low-level aesthetic (e.g., hue count) and high-level semantic visual features (e.g., food cues, branding) that were prevalent in a two-year comprehensive corpus of Facebook cannabis commercial posts. Multilevel ZINB models further revealed specific visual features that differentially predicted three engagement metrics. Regulatory and theoretical implications are discussed below.

Previous research on online cannabis product posts has often overlooked the rich visual data prevalent on social media. Even when visuals are considered, scalability remains a challenge since researchers typically rely on manual content analysis to process multimodal data (Lim et al., 2021; Olsson et al., 2025; Yang et al., 2018). We demonstrate that state-of-the-art computer vision techniques powered by deep learning can substantially enhance researchers’ toolkit for efficiently analyzing and monitoring visual appeals in online cannabis-promoting posts at scale. Our approach integrates unsupervised image clustering analysis with knowledge-guided visual feature extraction. This methodological eclecticism improves researcher flexibility in exploring rapidly evolving online cannabis marketing, particularly the creative use of various visual appeals, while maintaining measurement accuracy and validity for specific features of public health significance. This is especially important for the surveillance of potentially illegal visual appeals (e.g., youth-appealing content) under state-level regulations. Although our discussion focuses on visual appeals, multimodal deep learning models such as LLaVA and GPT-4o can also handle textual data, even those embedded in images. For example, we identified textual health benefits claims and warnings and included them as covariates in our regression analysis.

This methodological advance to process multimodal online cannabis marketing posts at scale is critical for informing regulatory policies and evaluating compliance. At present, many states (e.g., Illinois, Maine, Massachusetts) restrict therapeutic or curative claims in cannabis marketing (Hybrid Marketing Co., n.d.; Network for Public Health Law, 2022). Our unsupervised image clustering analysis identified posts making explicit or implicit (“healthy lifestyle”) therapeutic claims in text or images, indicating potential regulatory violations, though some CBD products may fall under different regulations. Additionally, some states like Illinois prohibit images of cannabis leaves and buds (Hybrid Marketing Co., n.d.). Our supervised analysis through the LLaVA model showed that 11.4% of posts contained “weed plant” images, showing significant non-compliance. Such large-scale auditing of specific visual marketing appeals and their regulatory compliance would be impractical without leveraging advanced computer vision techniques.

Furthermore, our computational analysis identified visual food cues as an understudied cannabis marketing tactic that warrants regulatory oversight. While most states prohibit advertising targeting those under 21 (Hybrid Marketing Co., n.d.), few have precisely defined what content features count as “youth-appealing” beyond banning “cartoons and toys” (e.g., Alaska, Montana, Nevada, New Jersey) and “candies” (e.g.,

Maine, Nevada). Our analysis revealed that visual food cues, including portrayals of candies, chocolates, fruits, and brownies, comprised 17.9% of posts and consistently drove higher engagement across all three metrics, suggesting strong viral marketing potential. Given that similar sensory appeals in tobacco products (e.g., fruity flavoring) attract adolescents (Baker et al., 2021; Chen-Sankey et al., 2024; Gaiha et al., 2022) and food cues are likely to trigger positive psychological and neural responses associated with food consumption (Rolls et al., 2016), visual food cues in online cannabis marketing warrant regulatory scrutiny.

Besides regulatory implications, sensory appeals like food cues merit scholarly attention in understanding audience engagement with online health advertising. While research shows emotional appeals drive message retransmission (Berger, 2014; Keib et al., 2018), specific content features constituting these appeals remain largely unexplored in the context of cannabis-related messages. Food cues’ known ability to activate brain reward systems (Rolls et al., 2016) suggests their visual depictions in cannabis marketing may generate arousal and positive emotions, potentially explaining their consistent association with engagement metrics. Although this emotional pathway awaits empirical verification in future research, our study emphasizes the need to consider visual sensory appeals like food cues in understanding audience engagement with online health marketing. Our study also reveals that semantic features (humans, animals) and aesthetic features show distinct associations with engagement metrics, emphasizing the need to unpack heterogeneity when examining visual features’ distinctive impact on engagement.

Beyond visual marketing strategies, our findings from RQ3 highlight a pronounced imbalance in health-related textual claims within the promotional posts. Nearly half of the posts mentioned health benefits, whereas information on health risks or safety concerns was rare and peripheral, and co-occurrence of benefits and risks was virtually absent. Qualitative thematic analysis further showed that health benefits claims predominantly centered on wellness, CBD-related therapeutic benefits, stress relief, sleep support, and pain relief. In contrast, disclosures regarding health risks and safety concerns were largely limited to regulatory disclaimers and responsible-use warnings, often embedded within promotional content and rendered visually inconspicuous. From a public health perspective, this asymmetry suggests that cannabis marketing on social media may disproportionately amplify health benefits, many lacking rigorous medical evidence, while inadequately disclosing potential risks and safety considerations, raising concerns about consumer misperceptions and youth exposure.

Under Meta’s Advertising Standards, ads may not promote or offer the sale of THC products or cannabis products containing psychoactive components, regardless of state legality (Meta, n.d.). Advertisements for CBD products are permitted only with prior written authorization and active certification, must comply with local laws and platform-specific targeting requirements, and may only target individuals aged 18 years or older in approved jurisdictions. Informational, advocacy, or public service content may be allowed provided it does not promote sales or consumption. However, our findings suggest that promotional cannabis-related content (e.g., particularly posts featuring sensory food cues and health-related framing) continues to circulate and generate substantial engagement within the platform ecosystem.

In light of these findings, scalable monitoring approaches can support regulatory oversight and public health surveillance in rapidly evolving digital marketing environments. As recreational cannabis legalization expands, identifying patterns of potentially youth-appealing content may assist stakeholders in assessing compliance with age-restriction policies, particularly given the imprecision surrounding what constitutes “youth-appealing” material in digital contexts. At the same time, AI-assisted monitoring systems require careful governance. Clear documentation, human oversight, and routine auditing are essential to ensure alignment with public health objectives and to prevent misuse in refining persuasive marketing strategies.

There are several limitations to consider. First, the scope of our data

is limited to Facebook posts, and social media platforms differ substantially in their algorithms, audience composition, and content affordances. These platform-specific dynamics may shape both marketing strategies and engagement patterns. Future research should extend this framework across multiple platforms and incorporate additional modalities, such as video-based content, to strengthen cross-platform generalizability. Second, while multimodal models like LLaVA are promising for monitoring online health marketing, their performance sensitivity to domain shifts and platform changes necessitates ongoing validation to maintain robustness. Changes in platform design, visual conventions, or marketing tactics could affect model robustness. Ongoing validation is therefore necessary, particularly when such systems are used in regulatory or public health contexts. Future work should also examine potential sources of algorithmic bias across cultural representations and demographic contexts to ensure equitable and accurate monitoring. Third, our reliance on aggregate engagement metrics (likes, comments, shares) limits insight into individual-level psychological processes. While these metrics capture observable interaction behavior, they do not reveal user motivations, risk perceptions, or behavioral outcomes. Experimental and survey-based studies are needed to clarify the mechanisms linking specific visual appeals to engagement and to assess how these effects may vary across demographic groups.

4. Conclusion

This study demonstrates the feasibility and value of using computer vision and multimodal deep learning techniques (image clustering algorithms, LLaVA, GPT-4o) to analyze visual features in online cannabis marketing at scale. Our findings reveal concerning patterns in the use of sensory (e.g., food cues) and other visual strategies, while identifying multiple instances of potential non-compliance with state-level regulations on online cannabis marketing. In particular, visual food cues were not only prevalent but consistently associated with higher levels of likes, comments, and shares, suggesting their strong engagement potential. Different semantic features (humans, animals) and color-related aesthetic features showed distinct associations with engagement metrics, highlighting the complex nature of audience engagement with online cannabis marketing.

Beyond documenting these patterns, our findings raise important regulatory and public health considerations. The prominence of youth-appealing sensory cues and the limited presence of risk information suggest an imbalance that may amplify exposure while minimizing awareness of potential harms. As recreational cannabis legalization expands, scalable monitoring approaches can support more evidence-

informed oversight of digital marketing practices, especially those that may disproportionately affect adolescents and young adults.

At the same time, the integration of AI-based monitoring tools into regulatory and research contexts requires careful governance. Transparent deployment, human oversight, and clear accountability structures are essential to ensure that such systems serve public health objectives rather than commercial optimization. When responsibly implemented, multimodal analytical tools offer meaningful potential to strengthen digital health surveillance and protect vulnerable populations in rapidly evolving online marketplaces.

Funding statement

This work was supported by the Office of the Vice Chancellor for Research and Graduate Education at the University of Wisconsin–Madison with funding from the Wisconsin Alumni Research Foundation to SY. We also received support from the Google Cloud Research Credits program awarded to SY. The research presented in this paper is that of the authors and does not reflect the opinions of the funding agency.

CRediT authorship contribution statement

Linqi Lu: Conceptualization, Data curation, Formal analysis, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Xianshi Yu:** Methodology, Writing – review & editing. **Zhuoyan Xu:** Methodology, Writing – review & editing. **Hyerin Kwon:** Methodology, Writing – review & editing. **Akhil P. Reddy:** Methodology, Writing – review & editing. **Haohang Xin:** Data curation, Writing – review & editing. **Shifan Zhang:** Data curation, Writing – review & editing. **Ellie Fan Yang:** Writing – original draft, Writing – review & editing. **Yin Li:** Conceptualization, Supervision, Writing – review & editing. **Sijia Yang:** Conceptualization, Funding acquisition, Supervision, Writing – review & editing.

Declaration of Competing Interest

All authors have no conflict of interest to disclose.

Acknowledgement

The research protocol was approved by the Institutional Review Board of the University of Wisconsin-Madison (ID: 2021–0616).

Appendix I. Multicollinearity Analysis

Table 1
Eigenvalues for Extracted Factors

Factor	Eigenvalue	% of Variance Explained	Cumulative % of Variance
Factor 1	2.08152296	8.005857538	8.005857538
Factor 2	1.60469753	6.171913577	14.17777112
Factor 3	1.45813655	5.6082175	19.78598862
Factor 4	1.31580073	5.060772038	24.84676065
Factor 5	1.25760765	4.8369525	29.68371315
Factor 6	1.16382897	4.476265269	34.15997842
Factor 7	1.09547645	4.213370962	38.37334938
Factor 8	1.06977205	4.114507885	42.48785727
Factor 9	1.04997445	4.038363269	46.52622054
Factor 10	1.0254546	3.944056154	50.47027669
Factor 11	1.00891979	3.880460731	54.35073742
Factor 12	1.00095165	3.849814038	58.20055146

Table 2
Rotated Factor Loadings with Varimax Rotation

	Factor 1	Factor 2	Factor 3	Factor 4
Visual Feature	Visual food cues	Health risk warning	Time feature	Time feature
edible	0.537	-0.034	-0.127	0.33
fruit	0.435	0.042	-0.044	0.234
bakery	0.425	0	0.04	-0.168
candy	0.466	0.016	-0.057	0.124
chocolate	0.564	-0.008	0.014	-0.072
weed plant	-0.086	0.053	0.011	-0.143
human presence	-0.087	-0.012	-0.014	-0.035
animal	-0.088	-0.022	-0.036	0.052
animation	-0.027	-0.001	0.01	-0.078
branding	0.095	-0.001	-0.072	0.283
warning label	0.04	0.662	0.002	0.036
Textual Feature				
call to action	-0.013	-0.068	0.046	0.064
health benefit	0.024	-0.119	-0.15	0.327
health risk	0.041	0.807	0.006	0.029
Covariates				
weekend	0.001	0.008	-0.003	-0.017
live video complete	-0.001	-0.002	0.002	-0.021
live video scheduled	-0.009	-0.002	-0.01	-0.032
native video	0.02	0.007	-0.002	-0.054
photo	0	0.001	-0.01	0.414
status	0.109	-0.011	0.024	-0.198
video	-0.009	-0.005	-0.006	-0.109
youtube link	-0.003	-0.008	-0.018	-0.118
evening	-0.02	-0.008	-0.214	0.079
morning	0.011	-0.054	0.995	0.131
night	0.013	0.057	-0.145	-0.167
subscriber number	-0.012	-0.003	0.004	-0.038

Note. A factor analysis with varimax rotation was conducted to examine the underlying structure of the visual and textual features. Five factors with eigenvalues greater than 1 were retained, accounting for 30.76% of the total variance (see Table 1). The first factor, 'Edible and Food Content,' included variables such as *edible*, *candy*, and *chocolate*, indicating that food-related posts tend to group together (see Table 2).

Appendix. II

Keyword List:

ListA: kief OR mmj OR mmot OR weed brownie OR weedbrownie OR weedbrownies OR weedbrownies OR Blunt OR budder OR Burning one down OR Butane honey oil OR Cannabis OR cannabis butter OR cannabis candy OR cannabis candy OR cannabis edible OR cannabis infused OR cannabutter OR Cbd OR Cheeching OR Chillum OR Concentrates OR Cosmic brownie OR Dab OR dab life OR dabber OR dabbers OR dabbin OR Dablife OR dabs OR dank cookie OR dank cookies OR dank edible OR edibles OR email OR e-nail OR ganja OR hash butter OR hash cookie OR hash oil OR hemp OR high edible OR indica OR infused marijuana OR kush OR kush butter OR kush cookies OR kush infused OR Legalize marijuana OR legalized marijuana OR legalizing marijuana OR Live resin OR magic brownie OR magic brownies OR Marihuana OR Marijuana OR Marijuana brownie OR marijuana butter OR Marijuana cake OR Marijuana candy OR marijuana cookie OR marijuana edible OR Marijuana gummies OR medibles OR Medical cannabis

ListB: Medical marijuana OR Medicated brownie OR Medicated cake OR Medicated candy OR Medicated cookie OR Medicated gummies OR Need blunt OR need pot OR Poking OR Pot brownie OR pot butter OR pot candy OR Pot cookie OR pot edible OR Pot gummies OR pot infused OR Pothead OR Recreational marijuana OR reefer OR Rosin OR sativa OR shatter day OR Shatterday OR Smoke blunt OR Smoke mary jane OR smoke pot OR smoke shatter OR smoking shatter OR Smoking trees OR Solventless OR Space brownie OR Space cake OR Space candy OR Special brownie OR Special gummies OR Spliff OR Stoned OR Stoner OR the OR the butter OR the candy OR the cookie OR the cookie OR the edible OR Toking OR Want a blunt OR want a joint OR want pot OR Wax pen OR Wax pens OR Weed OR Weed brownie OR weed butter OR Weed cake OR Weed candy OR weed edible OR weed gummies OR weed infused OR weed-infused

Appendix. III Classification Prompts

Label	Description	Prompt
Candy	Determine if the product is candy.	0: No candy present. 1: Candy is present.
Bakery	Determine if the product is a bakery such as a cookie, brownie, cake, etc.	0: No bakery items present. 1: Bakery item is present.
Chocolate	Determine if the product is chocolate.	0: No chocolate present. 1: Chocolate is present.
Fruit	Determine if the product is fruit or contains fruit.	0: No fruit present. 1: Fruit is present.
Weed Plant	Identify if there is a weed (cannabis/marijuana) plant present.	0: No weed plant present. 1: Weed plant is present.
Animation	Identify whether the content is animated (cartoons, animated characters, etc.).	0: No animation. 1: Animation detected.

(continued on next page)

(continued)

Label	Description	Prompt
Animal	Identify if a real animal is present in the image and provide a description.	0: No animal presence. 1: Animal presence detected. (Please describe the detected animal)
Branding	Determine if there is visible branding, such as logos or branded packaging.	0: No visible branding. 1: Visible branding present.
Human Presence	Identify whether a human is present in the image or description.	0: No human presence. 1: Human presence detected.
Interactive Features	Detect whether the image includes interactive elements designed to engage the user, such as QR codes, clickable buttons, links to social media, or gamification features like polls or quizzes.	0: No interactive features are present. 1: Interactive features are present.
Commercial Post	Detect whether a post is likely to be a marijuana-related commercial. This would typically involve evaluating keywords, phrases, and contexts that are commonly associated with commercial activity, such as promotions, sales language, product mentions, or service offerings.	0: Non-commercial post 1: Commercial post
Health Claims and Warnings	Detect whether the post includes any health-related messaging. This includes both warnings about potential health risks and claims about health benefits. Look for any statements that make assertions about health impacts, whether positive or negative.	0: No health-related claims are present in the post. 1: Health-related benefit claims are present in the post. 2: Health-related risk claims are present in the post.

References

Allem, J.-P., Escobedo, P., Dharmapuri, L., 2020. Cannabis surveillance with Twitter data: emerging topics and social bots. *Am. J. Public Health* 110 (3), 357–362. <https://doi.org/10.2105/AJPH.2019.305461>.

Atkins, D.C., Baldwin, S.A., Zheng, C., Gallop, R.J., Neighbors, C., 2013. A tutorial on count regression and zero-altered count models for longitudinal substance use data. *Psychol. Addict. Behav.* 27 (1), 166–177. <https://doi.org/10.1037/a0029508>.

Baker, A.N., Wilson, S.J., Hayes, J.E., 2021. Flavor and product messaging are the two most important drivers of electronic cigarette selection in a choice-based task. *Sci. Rep.* 11, 4689. <https://doi.org/10.1038/s41598-021-83796-5>.

Bakshy, E., Messing, S., Adamic, L.A., 2015. Exposure to ideologically diverse news and opinion on Facebook. *Science* 348 (6239), 1130–1132. <https://doi.org/10.1126/science.aaa1160>.

Berger, J., 2014. Word of mouth and interpersonal communication: A review and directions for future research. *J. Consum. Psychol.* 24 (4), 586–607. <https://doi.org/10.1016/j.jcps.2014.05.002>.

Cappella, J.N., Kim, H.S., Albarracín, D., 2014. Selection and transmission processes for information in the emerging media environment: Psychological motives and message characteristics. *Media Psychol.* 18 (3), 396–424. <https://doi.org/10.1080/15213269.2014.941112>.

Cavazos-Rehg, P.A., Krauss, M.J., Cahn, E., Lee, K.E., Ferguson, E., Rajbhandari, B., Sowles, S.J., Floyd, G.M., Berg, C., Bierut, L.J., 2019. Marijuana promotion online: An investigation of dispensary practices. *Prev. Sci.* 20 (2), 280–290. <https://doi.org/10.1007/s11221-018-0889-2>.

Chen-Sankey, J., Jeong, M., Wackowski, O.A., Unger, J.B., Niederdeppe, J., Bernat, E., Choi, K., 2024. Noticing people, discounts, and non-tobacco flavors in e-cigarette ads may increase e-cigarette product appeal among non-tobacco-using young adults. *Tob. Control* 33 (1), 30–37. <https://doi.org/10.1136/tobaccocontrol-2022-057756>.

Cui, Y., Duan, Z., LoParco, C.R., Vinson, K., Romm, K.F., Wang, Y., Cavazos-Rehg, P.A., Kasson, E., Yang, Y.T., Berg, C.J., 2024. Changes in online marketing and sales practices among non-medical cannabis retailers in 5 US cities, 2022–2023. *Prev. Med. Rep.* 42, 102755. <https://doi.org/10.1016/j.pmedr.2024.102755>.

Deng, T., Urbaczewski, A., Lee, Y.J., Barman-Adhikari, A., Dewri, R., 2024. Identifying Marijuana Use Behaviors Among Youth Experiencing Homelessness Using a Machine Learning-Based Framework: Development and Evaluation Study. *JMIR AI* 3 (1), e53488.

Gaiha, S.M., Lempert, L.K., McKelvey, K., Halpern-Felsher, B., 2022. E-cigarette devices, brands, and flavors attract youth: Informing FDA's policies and priorities to close critical gaps. *Addict. Behav.* 126, 107179. <https://doi.org/10.1016/j.addbeh.2021.107179>.

Goldberg, M.E., Gorn, G.J., 1987. Happy and sad TV programs: How they affect reactions to commercials. *J. Consum. Res.* 14, 387. <https://doi.org/10.1086/209122>.

Grimmer, J., Roberts, M.E., Stewart, B.M., 2021. Machine learning for social science. *Annu. Rev. Political Sci.* 24, 395–419. <https://doi.org/10.1146/annurev-polisci-053119-015921>.

Hasler, D., Suesstrunk, S.E., 2003. June. Measuring colorfulness in natural images. Vol. 5007. In: *Human vision and electronic imaging, VIII. SPIE*, pp. 87–95. Vol. 5007.

Hilbe, J.M., 2011. *Negative binomial regression*, 2nd ed. Cambridge University Press.

Holbrook, M.B., Batra, R., 1987. Assessing the role of emotions as mediators of consumer responses to advertising. *J. Consumer. Res.* 14 (3), 404–420.

Hu, M., Benson, R., Chen, A.T., Zhu, S.H., Conway, M., 2021. Determining the prevalence of cannabis, tobacco, and vaping device mentions in online communities using natural language processing. *Drug Alcohol Depend.* 228, 109016.

Hybrid Marketing Co. (n.d.). *Cannabis marketing guidelines by state*. Retrieved March 6, 2025, from (<https://hybridmarketingco.com/cannabis-marketing-guidelines-by-stat-e/>)

Ishida, J.H., Zhang, A.J., Steigerwald, S., Cohen, B.E., Vali, M., Keyhani, S., 2020. Sources of information and beliefs about the health effects of marijuana. *J. Gen. Intern. Med.* 35 (1), 153–159. <https://doi.org/10.1007/s11606-019-05335-6>.

Johnston, L.D., Miech, R.A., O'Malley, P.M., Bachman, J.G., Schulenberg, J.E., Patrick, M.E., 2020. *Monitoring the Future National Survey Results on Drug Use, 1975–2019: Overview, key findings on adolescent drug use*. Institute for Social Research, University of Michigan.

Keib, K., Espina, C., Lee, Y.-I., Wojdyski, B.W., Choi, D., Bang, H., 2018. Picture this: The influence of emotionally valenced images on attention, selection, and sharing of social media news. *Media Psychol.* 21 (2), 202–221. <https://doi.org/10.1080/15213269.2017.1378108>.

Ketchen, D.J., Shook, C.L., 1996. The application of cluster analysis in strategic management research: an analysis and critique. *Strateg. Manag. J.* 17 (6), 441–458.

Krauss, M.J., Sowles, S.J., Sehi, A., Spitznagel, E.L., Berg, C.J., Bierut, L.J., Cavazos-Rehg, P.A., 2017. Marijuana advertising exposure among current marijuana users in the U.S. *Drug Alcohol Depend.* 174, 192–200. <https://doi.org/10.1016/j.drugalcdep.2017.01.017>.

Lang, P.J., Greenwald, M.K., Bradley, M.M., Hamm, A.O., 1993. Looking at pictures: affective, facial, visceral, and behavioral reactions. *Psychophysiology* 30 (3), 261–273. <https://doi.org/10.1111/j.1469-8986.1993.tb03352.x>.

Lim, C.C., Leung, J., Chung, J.Y.C., Sun, T., Gartner, C., Connor, J., Chan, G., 2021. Content analysis of cannabis vaping videos on YouTube. *Addiction* 116 (9), 2443–2453.

Liu, H., Li, C., Wu, Q., Lee, Y.J., 2023. Visual instruction tuning. *Adv. Neural Inf. Process. Syst.* 36. <https://arxiv.org/abs/2304.08485>.

Lu, L., Kwon, H., Wang, W., M. Cotter, L., A. Kriss, L., Minich, M., Liu, J., Cascio, C.N., Yang, S., 2025. Cannabis warning labels, sensory marketing, and electronic word-of-mouth: AI-facilitated textual analysis of a randomized experiment among youth and young adults. *Int. J. Advert.* 1–26. <https://doi.org/10.1080/02650487.2025.2548647>.

Luc, M.H., Tsang, S.W., Thrul, J., Kennedy, R.D., Moran, M.B., 2020. Content analysis of online product descriptions from cannabis retailers in six US states. *Int. J. Drug Policy* 75, 102593. <https://doi.org/10.1016/j.drugpo.2019.10.017>.

MacQueen, J., 1967. Some methods for classification and analysis of multivariate observations. *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*. University of California Press, pp. 281–297.

Meta. (n.d.). *Drugs and pharmaceuticals*. Retrieved March 3, 2026, from (<https://transparency.meta.com/policies/ad-standards/restricted-goods-services/drugs-pharmaceuticals/>)

Moreno, M.A., Gower, A.D., Jenkins, M.C., Kerr, B., Gritton, J., Gunther, J., 2019. Engagement with marijuana marketing on social media among U.S. adolescents and emerging adults. *Drug Alcohol Depend.* 204, 107585. <https://doi.org/10.1016/j.drugalcdep.2019.05.019>.

National Conference of State Legislatures, 2024. *Cannabis Overv.* (<https://www.ncsl.org/civil-and-criminal-justice/cannabis-overview/>).

Network for Public Health Law, 2022. *State regulation of adult-use cannabis advertising*. Retrieved from (<https://www.networkforphl.org/wp-content/uploads/2022/11/State-Regulation-of-Adult-Use-Cannabis-Advertising.pdf>). Retrieved from (<https://www.networkforphl.org/wp-content/uploads/2022/11/State-Regulation-of-Adult-Use-Cannabis-Advertising.pdf>).

Olsson, S.E., Sekhon, V.K., LoParco, C.R., Yockey, R.A., Greene, K.M., Henry, D., Rossheim, M.E., 2025. Portrayal of Delta-8 Tetrahydrocannabinol (THC) on YouTube. *Subst. Use & Misuse* 1–7.

Peng, Y., 2022. *Athec: A Python library for computational aesthetic analysis of visual media in social science research*. *Comput. Commun. Res.* 4 (1).

Peng, Y., Lock, I., Ali Salah, A., 2024. Automated visual analysis for the study of social media effects: Opportunities, approaches, and challenges. *Commun. Methods Meas.* 18 (2), 163–185.

- Pew Research Center, 2018. Teens, social media & technology 2018. (<https://www.pewresearch.org/internet/2018/05/31/teens-social-media-technology-2018/>).
- Pieters, R., Wedel, M., Batra, R., 2010. The stopping power of advertising: Measures and effects of visual complexity. *J. Mark.* 74 (5), 48–60. <https://doi.org/10.1509/jmkg.74.5.048>.
- Primack, B.A., Colditz, J.B., Pang, K.C., Jackson, K.M., Cotter, T., Kraemer, K.L., 2017. Important health topics on Twitter: A systematic review. *Am. J. Health Promot.* 31 (3), 218–230. <https://doi.org/10.4278/ajhp.141207-LIT-605>.
- Radford, A., Learning transferable visual models from natural language supervision. In: *Proceedings of the 38th International Conference on Machine Learning*. PMLR; 2021: 8748–8763.
- Raudenbush, S.W., Bryk, A.S., 2002. *Hierarchical Linear Models: Applications and Data Analysis Methods*, 2nd ed. Sage Publications, Thousand Oaks, CA.
- Rolls, E.T., 2016. Reward systems in the brain and nutrition. *Annu. Rev. Nutr.* 36 (1), 435–470. <https://doi.org/10.1146/annurev-nutr-071715-050936>.
- Simonyan, K., Zisserman, A., 2014. Very Deep convolutional Netw. LargeScale Image Recognit. *arXiv Prepr. arXiv:1409.1556*.
- Spillane, T.E., Wong, B.A., Giovenco, D.P., 2021. Content analysis of instagram posts by leading cannabis vaporizer brands. *Drug Alcohol Depend.* 218, 108353. <https://doi.org/10.1016/j.drugalcdep.2020.108353>.
- Thorson, K., Wells, C., 2016. Curated flows: A framework for mapping media exposure in the digital age. *Commun. Theory* 26 (3), 309–328. <https://doi.org/10.1111/comt.12087>.
- Trangenstein, P.J., Whitehill, J.M., Jenkins, M.C., Jernigan, D.H., Moreno, M.A., 2019. Active cannabis marketing and adolescent past-year cannabis use. *Drug Alcohol Depend.* 204, 107548. <https://doi.org/10.1016/j.drugalcdep.2019.107548>.
- Trangenstein, P.J., Gora Combs, K., Whitehill, J.M., Jenkins, M.C., Jernigan, D.H., Moreno, M.A., 2022. Typology of adolescents exposed to non-medical cannabis marketing and associations with consumption patterns. *Prev. Sci.* 23 (7), 1276–1286. <https://doi.org/10.1007/s11121-022-01378-0>.
- Yang, Q., Sangalang, A., Rooney, M., Maloney, E., Emery, S., Cappella, J.N., 2018. How is marijuana vaping portrayed on YouTube? Content, features, popularity and retransmission of vaping marijuana YouTube videos. *J. Health Commun.* 23 (4), 360–369.
- Yuan, Y., Kasson, E., Taylor, J., Cavazos-Rehg, P., De Choudhury, M., Aledavood, T., 2024. Examining the gateway hypothesis and mapping substance use pathways on social media: machine learning approach. *JMIR Form. Res.* 8, e54433.